

Towhid Ahmed

Generative AI Engineer · Open to remote — worldwide

towhidahmed.eng@gmail.com | GitHub | LinkedIn | Portfolio | +880 1871 834-635

Professional Summary

Generative AI Engineer with 1+ year of freelance experience building production-grade LLM applications, multi-agent systems, and RAG solutions for SaaS products. Developed 3 AI applications using LangChain, LangGraph, and FastAPI, achieving 35% hallucination reduction, 25% retrieval relevance improvement, and 98/98 passing automated tests. Experienced in RAG pipelines, LLMops, Docker, and LangSmith.

Technical Skills

Programming: Python, SQL

Generative AI: LLMs, RAG, Advanced RAG, AI Agents, Prompt Engineering, LangChain, LangGraph

LLMs & Models: OpenAI, Llama, Groq, Hugging Face Transformers, Sentence Transformers

Databases: Pinecone, ChromaDB, PostgreSQL

Backend & Deployment: FastAPI, Docker, Git, GitHub Actions, CI/CD

Testing & Observability: Pytest, LangSmith, LLM Guardrails

Software Engineering: Data Structures & Algorithms (400+ problems), OOP, LLM System Design

Professional Experience

Freelance Generative AI Engineer | Self-Employed | Jun 2025 – Present

AI Document Assistant | Jun 2026 – Jul 2026

- Enhanced information reliability and user trust by developing a **3-agent LangGraph workflow** with **citation-based RAG**, significantly **reducing AI hallucinations**.
- Powered seamless document processing and interaction by engineering **7 production-ready REST APIs** using **FastAPI**, **OpenAI API**, and **Pinecone**.
- Enabled **1,000+** document knowledge-base updates without redeployment.

AI Sales & Customer Support Assistant | Aug 2025 – Sep 2025

- **Automated customer interactions and improved consistency** using a multi-agent **LangGraph pipeline** (Intent, Retrieval, Response) with Pinecone RAG and **conversational memory**.
- Slashed AI inference costs and maintained high execution speed by integrating **Groq**, Hugging Face embeddings, and **Pinecone Serverless**.
- **Built a scalable support infrastructure** with a Streamlit UI and a FastAPI backend featuring **6+ REST APIs** for knowledge uploads and session management.

Selected Projects

Multi-Agent AI System — Python, LangGraph, LangChain, FastAPI, Pinecone, LangSmith, Streamlit, Docker, Render, Hugging Face, Tavily GitHub | Demo

- **Designed 4-agent LangGraph workflow** routing queries between **RAG** and **web search** based on intent and complexity.
- Built **FastAPI** backend with async execution, **7 core APIs** (/chat, /upload, /health, /metrics) optimized for **Render**.
- Integrated **Tavily** with **DuckDuckGo fallback** and **LangSmith tracing** for reliable web retrieval and **observability**.

Multimodal RAG — Python, FastAPI, Streamlit, Pinecone, Groq, Llama 3.3, Faster-Whisper, RAG, Vector Search, LangSmith, Docker, Git, GitHub, REST API, Async Programming, Caching GitHub

- Designed production multimodal retrieval (PDF/audio/video) **improving relevance by 25%** and **reducing hallucinations by 35%**.
- Implemented two-stage retrieval (**semantic chunking** + cross-encoder **re-ranking**) with **<2s latency** via **caching**.
- Tech: **Groq Llama 3.3**, **BAAI embeddings**, **Pinecone**, **FastAPI**, **Streamlit**, **Docker**; 76-unit test coverage.

Clinical RAG Assistant — Python, FastAPI, LangGraph, RAG, Pinecone, LLM Guardrails, Tavily, Streamlit, LangSmith, Docker, JWT GitHub

- Built a clinical information assistant using a **4-agent LangGraph pipeline** (Router, Safety, Retrieval, Answer) with **RAG and web search**.
- Implemented **multi-layer safety system** (input filters, retrieval validation, output sanitization) with JWT auth.
- Achieved **98/98** passing tests across **guardrails**, **RAG evaluation**, and **API workflows**.

Education and Certificates

Education: Computer Science Engineering(CSE), Chittagong University of Engineering & Technology | 2023-2027

Certifications: • Generative AI & Agentic AI — Krish Naik (Hands-on LLM systems & agents)

• LLMops — Udemy (Deployment, monitoring, CI/CD for LLM apps) • Data Structures & Algorithms — TakeUForward